Cantera vs neural networks | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 04 | NRMSE (util.py)

T0 = 1317.5 K P0 = 1.244 atm phi0 = 1.984 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0108	0.0583	0.593	2.33	0.0953	0.832	0.0354	0.149	0.418	0	Complete
2 P-only DeepONet10	**10k**	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	Complete
2 P-only DeepONet10	**long**	2.04e-03	1.92e-03	0.0153	0.0608	0.0293	0.0552	2.39e-03	0.0633	0.0794	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	9.87e-03	0.0137	0.196	0.258	0.0434	0.031	2.92e-03	0.155	0.2	0	Complete
3 ANN10 + P-only DeepONet10	**long**	7.36e-04	1.14e-03	0.0116	0.026	8.30e-03	0.0299	1.04e-03	0.0362	0.0393	0	Complete
4 T/P DeepONet10	**10k**	0.0189	0.0104	0.227	0.392	0.0858	0.218	9.30e-03	0.336	0.28	0	Complete
4 T/P DeepONet10	**long**	0.154	0.0899	1.66	5.28	0.0687	3.15	0.111	0.543	105	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.016	8.47e-03	0.186	0.267	0.0804	0.21	8.43e-03	0.233	0.181	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.111	0.0576	0.958	2.81	0.0298	3.18	0.105	0.116	0.589	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0714	0.0485	0.943	0.811	0.196	0.973	0.0223	2.13	23.7	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.116	0.0915	1.74	1.93	0.0987	0.735	0.0284	0.368	2.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	4.70e-03	3.41e-03	0.059	0.0529	0.0354	0.0625	2.65e-03	0.16	0.106	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.101	0.0863	1.57	1.33	0.0648	0.735	0.0245	0.289	1.82	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 05 | NRMSE (util.py)

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.299	0.773	1.07	1.02	0.614	11.4	0.509	1.03	1.01	0	Complete
2 P-only DeepONet10	**10k**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
2 P-only DeepONet10	**long**	0.6	1.43	1.89	8.35	0.827	0.517	0.985	5.68	0.971	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.271	0.75	18.7	0.82	0.466	1.21	0.49	0.985	1	0	Complete
4 T/P DeepONet10	**10k**	0.35	0.942	2.92	0.887	0.604	1.32	0.626	0.989	0.997	0	Complete
4 T/P DeepONet10	**long**	0.547	0.964	53.1	0.734	0.939	25.1	0.669	1.01	1	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.34	0.913	2.09	0.728	0.586	1.34	0.606	0.921	0.996	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.375	0.893	16.8	0.975	0.819	18.8	0.566	0.894	0.992	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.427	0.877	68.1	0.592	0.648	13.1	0.551	7.69	0.966	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.305	0.885	30.5	0.776	0.627	8.42	0.521	2.09	0.991	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 07 | NRMSE (util.py)

T0 = 930.6 K P0 = 3.722 atm phi0 = 0.754 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	2.26e-06	1.72e-06	1	360	2.48e-06	1	1	0.531	0.464	0	Complete
2 P-only DeepONet10	**10k**	2.84e-06	2.61e-06	1	0.68	4.31e-06	0.986	0.999	0.994	1	0	Complete
2 P-only DeepONet10	**long**	1.12	0.631	2.57e+06	2.60e+08	0.494	1.03e+09	7.18e+05	65.3	782	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	2.84e-06	2.61e-06	1	0.623	4.31e-06	0.987	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	1.42	0.799	1.48e+06	3.29e+08	0.625	1.26e+09	9.10e+05	117	1.44e+03	0	Complete
4 T/P DeepONet10	**10k**	2.85e-06	2.61e-06	29.5	0.994	4.31e-06	1.56	1	0.993	1	0	Complete
4 T/P DeepONet10	**long**	0.307	0.975	6.25e+08	1.28e+08	0.969	3.29e+10	5.42e+05	39.2	886	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	2.86e-06	2.49e-06	105	0.972	4.01e-06	14.1	0.99	0.861	0.984	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.409	0.116	6.85e+08	3.71e+05	5.26e-03	1.56e+07	4.57e+03	146	2.09e+03	0	Complete
6 T/P/ϕ DeepONet10	**10k**	1.48	0.931	5.41e+07	3.02e+08	0.768	6.67e+09	9.77e+05	107	599	0	Complete
6 T/P/ϕ DeepONet10	**long**	7.51e-06	1.68e-06	9.73e+03	0.602	4.05e-06	323	0.798	0.945	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	1.38	0.888	5.23e+07	2.11e+08	0.74	7.44e+09	9.16e+05	119	900	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.0301	0.0167	102	650	0.0155	290	1.54e+04	176	3.15e+03	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

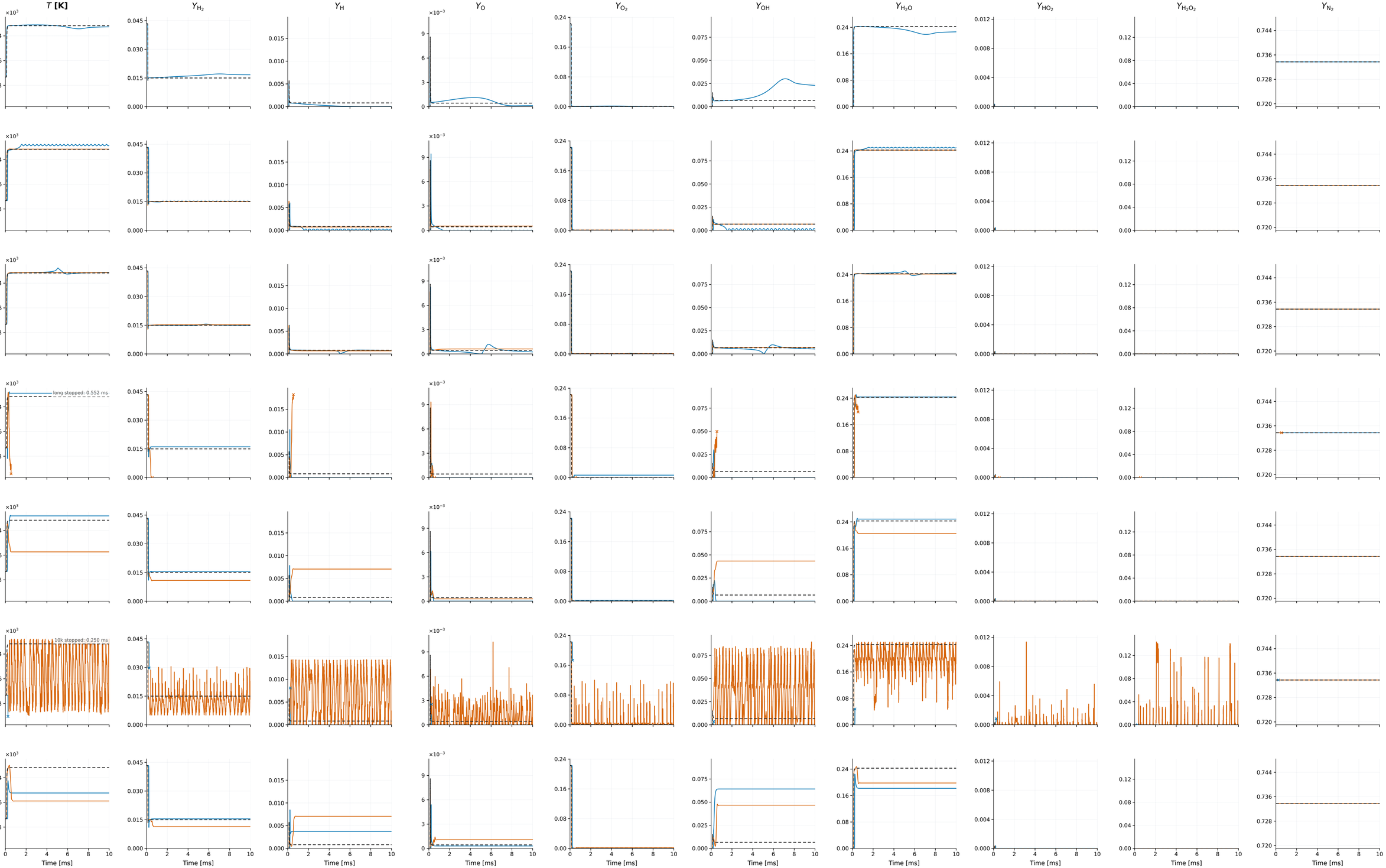
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 18 | NRMSE (util.py)

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0255	0.115	0.793	0.959	0.336	1.94	0.0671	1.02	1.19	0	Complete
2 P-only DeepONet10	**10k**	0.067	0.161	0.806	1.05	0.768	0.82	0.0892	1.54	2.36	0	Complete
2 P-only DeepONet10	**long**	7.36e-03	0.0402	0.184	0.374	0.17	0.0525	0.018	0.468	0.667	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.029	0.0992	0.314	0.756	0.462	0.297	0.051	1.31	1.86	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0118	0.0562	0.249	0.529	0.241	0.0846	0.0261	0.698	0.953	0	Complete
4 T/P DeepONet10	**10k**	0.0652	0.169	1.06	1.06	0.778	1.03	0.0828	1.98	5.67	0	Complete
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	**Stop 0.552 ms**
5 ANN10 + T/P DeepONet10	**10k**	0.0681	0.149	1.03	1.02	0.693	1.02	0.0796	1.62	2.2	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.371	0.261	7.26	0.43	0.129	5.42	0.155	0.327	0.576	0	Complete
6 T/P/ϕ DeepONet10	**10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.250 ms**
6 T/P/ϕ DeepONet10	**long**	0.464	0.331	8.77	2.73	0.568	5.54	0.235	27.6	2.16e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.304	0.175	3.44	0.629	0.833	8.52	0.265	2.04	5.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.389	0.223	7.2	1.26	0.0564	5.87	0.181	0.25	0.299	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 32 | NRMSE (util.py)

T0 = 1140.7 K P0 = 3.354 atm phi0 = 1.463 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0312	0.126	1.48	0.511	0.0533	0.83	0.0317	0.118	1.03	0	Complete
2 P-only DeepONet10	**10k**	0.0629	0.063	0.97	0.99	0.447	0.972	0.0492	1.24	1.51	0	Complete
2 P-only DeepONet10	**long**	2.20e-03	0.0192	0.0875	0.165	0.12	0.0203	6.87e-03	0.28	0.242	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	8.62e-03	0.0401	0.15	0.275	0.276	0.0993	0.0162	0.778	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	1.78e-03	0.0124	0.107	0.166	0.0455	0.027	3.34e-03	0.103	0.0928	0	Complete
4 T/P DeepONet10	**10k**	0.04	0.127	0.968	1	0.844	0.993	0.0346	2.01	13	0	Complete
4 T/P DeepONet10	**long**	0.46	0.432	9.77	0.918	0.202	4.85	0.174	0.81	31	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0538	0.0417	0.965	0.912	0.184	0.98	0.0292	0.891	0.997	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.043	0.0234	0.653	0.243	0.139	0.883	0.0315	0.353	0.361	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.285	0.0903	3.13	1.31	0.345	6.75	0.254	2.61	7.38	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.699	0.483	13.2	2.15	0.142	8.55	0.319	28.5	541	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.272	0.13	2.39	0.126	0.0193	7.65	0.272	1.92	9.92	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.266	0.0204	3.64	0.0449	0.0177	5.71	0.203	1.04	1.61	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 39 | NRMSE (util.py)

T0 = 1250.1 K P0 = 2.297 atm phi0 = 0.646 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.15	0.476	2.07	0.897	0.329	5.49	0.241	0.78	0.837	0	Complete
2 P-only DeepONet10	**10k**	3.93e-03	0.236	0.758	1.17	6.65e-03	0.321	6.61e-03	0.331	0.24	0	Complete
2 P-only DeepONet10	**long**	2.45e-04	0.0148	0.0609	0.0128	3.33e-03	0.0267	2.87e-04	0.0461	0.0488	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	5.60e-03	0.233	0.703	1.3	7.30e-03	0.33	6.99e-03	0.292	0.178	0	Complete
3 ANN10 + P-only DeepONet10	**long**	6.34e-04	0.0365	0.225	0.0833	3.70e-03	0.0233	1.55e-03	0.0378	0.0397	0	Complete
4 T/P DeepONet10	**10k**	0.0406	0.212	0.694	0.675	0.0647	0.897	0.0488	0.769	0.794	0	Complete
4 T/P DeepONet10	**long**	0.025	0.0416	0.369	0.0843	0.0523	0.77	0.034	0.205	0.226	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0362	0.13	0.568	0.562	0.0688	0.891	0.0437	0.833	0.881	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.339	0.38	11.5	0.435	0.582	9.41	0.42	0.91	0.968	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0408	0.197	0.787	0.562	0.0679	0.955	0.0499	0.634	0.567	0	Complete
6 T/P/ϕ DeepONet10	**long**	4.36e-04	0.0177	0.0281	0.0177	1.90e-03	0.0109	8.75e-04	0.0171	0.0157	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0351	0.121	0.697	0.309	0.0652	0.934	0.0446	0.686	0.692	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	3.77e-04	0.0501	0.0923	0.119	8.84e-03	0.106	1.76e-03	0.0227	0.0787	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

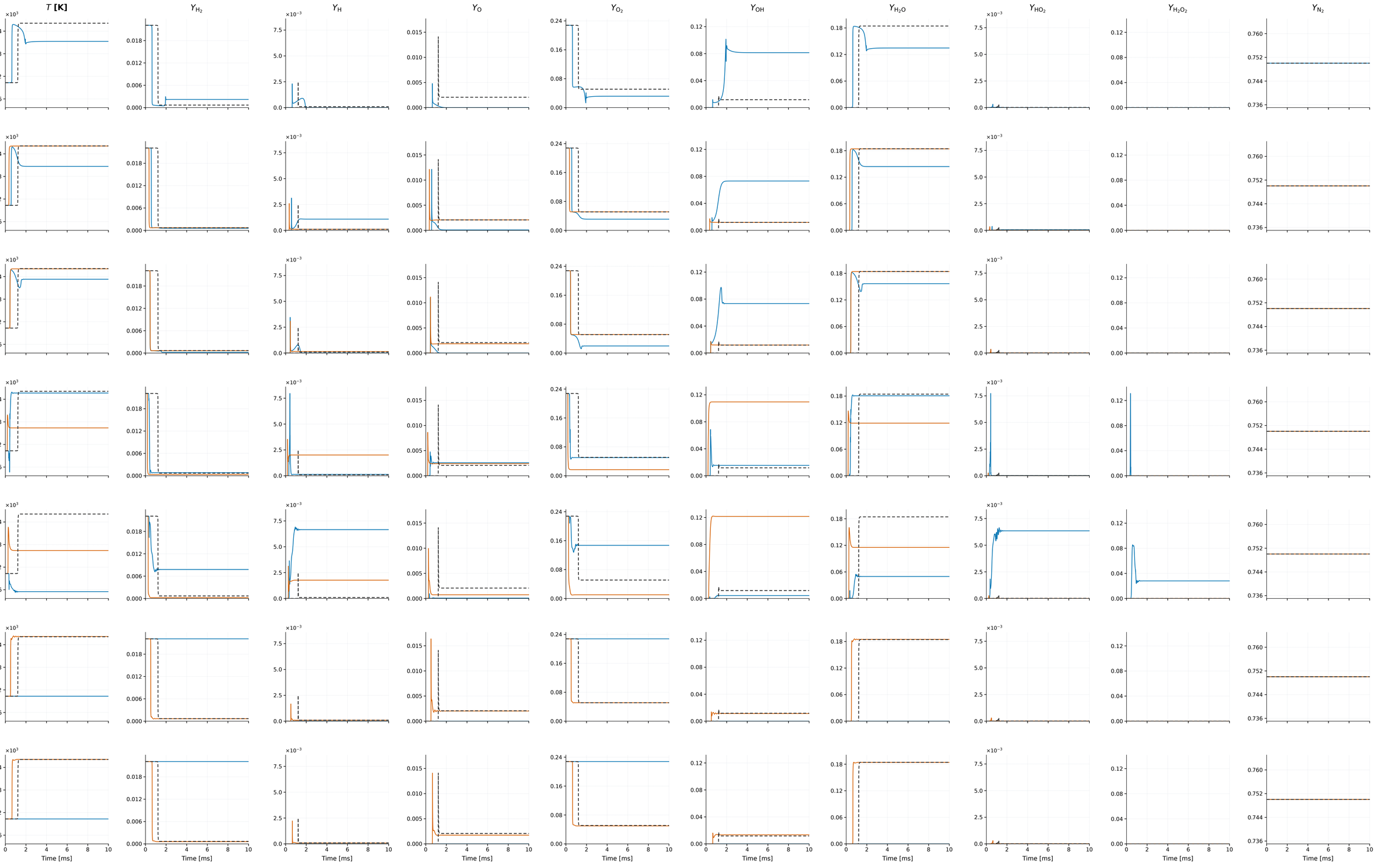
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 44 | NRMSE (util.py)

T0 = 1031.1 K P0 = 2.272 atm phi0 = 0.770 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.231	0.688	2.26	0.999	0.481	5.74	0.363	1.11	1.23	0	Complete
2 P-only DeepONet10	**10k**	0.252	0.701	8.52	0.997	0.528	5.19	0.335	1.82	1.38	0	Complete
2 P-only DeepONet10	**long**	0.186	0.813	0.878	0.4	0.559	0.324	0.312	1.03	1.03	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.194	0.773	1.77	1.02	0.628	5.32	0.317	1.03	0.986	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.174	0.763	1.02	0.379	0.524	0.306	0.292	1.07	1.08	0	Complete
4 T/P DeepONet10	**10k**	0.169	0.784	4.31	0.488	0.547	0.67	0.288	10.8	819	0	Complete
4 T/P DeepONet10	**long**	0.377	0.9	17.3	0.496	0.8	8.83	0.419	1.08	1.01	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.782	0.972	57.9	0.965	1	0.638	0.733	221	3.74e+03	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.377	0.871	15	0.721	0.821	9.83	0.432	1.1	1.03	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.168	0.731	0.687	0.447	0.503	0.276	0.281	1.1	1.19	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.149	0.654	0.729	0.423	0.452	0.311	0.251	1.08	1.24	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 45 | NRMSE (util.py)

T0 = 1181.9 K P0 = 4.842 atm phi0 = 1.099 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	8.22e-03	0.379	1.98	1.07	0.214	0.942	0.0629	1.61	5.76	0	Complete
2 P-only DeepONet10	**10k**	0.25	1.59	0.904	5.59	2.08	2.48	0.364	2	7.4	0	Complete
2 P-only DeepONet10	**long**	0.204	1.41	0.962	5.54	2.02	1.61	0.301	0.138	0.592	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.148	0.982	0.896	1.99	1.19	2.01	0.236	2.46	12.9	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0104	0.169	0.394	0.29	0.27	0.144	0.0281	0.433	0.46	0	Complete
4 T/P DeepONet10	**10k**	0.137	1.06	1.85	0.835	2.96	1.05	0.199	4.01	5.37	0	Complete
4 T/P DeepONet10	**long**	0.359	0.195	5.55	0.676	0.497	5.43	0.334	5.79	4.53	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0687	0.168	0.958	0.974	0.537	0.996	0.0649	1.19	0.682	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.302	0.443	3.09	0.817	0.587	5.53	0.334	4.18	6	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0835	0.76	0.851	1.55	2.22	0.889	0.134	2.86	5.55	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.238	0.193	3.42	0.516	0.299	3.76	0.234	3.01	2.99	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0594	0.181	0.981	0.969	0.723	0.997	0.0499	0.892	0.616	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	6.98e-03	0.0308	0.0829	0.0481	0.0741	0.134	8.99e-03	0.135	0.296	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

long stopped: 0.486 ms

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 53 | NRMSE (util.py)

T0 = 1257.8 K P0 = 2.103 atm phi0 = 1.799 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0249	0.105	0.84	0.642	0.0369	3.25	0.0957	0.118	0.566	0	Complete
2 P-only DeepONet10	**10k**	0.0551	0.0218	0.754	0.845	0.0818	0.868	0.03	0.247	0.641	0	Complete
2 P-only DeepONet10	**long**	6.02e-03	9.36e-03	0.124	0.107	0.0969	0.04	3.93e-03	0.253	0.156	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	2.57e-03	2.74e-03	0.0432	0.0484	0.0322	0.0285	1.59e-03	0.131	0.184	0	Complete
3 ANN10 + P-only DeepONet10	**long**	5.73e-03	9.72e-03	0.138	0.182	0.0662	0.0114	2.73e-03	0.158	0.109	0	Complete
4 T/P DeepONet10	**10k**	0.0311	0.0186	0.439	0.467	0.196	0.463	0.0176	0.488	0.557	0	Complete
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	**Stop 0.486 ms**
5 ANN10 + T/P DeepONet10	**10k**	0.0172	0.0109	0.236	0.198	0.13	0.287	0.0107	0.341	0.331	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0321	0.0129	0.439	0.247	0.0231	0.558	0.0178	0.139	0.34	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.149	0.0297	1.67	1.47	0.293	3.28	0.106	1.41	6.57	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.653	0.451	10.4	0.707	0.052	6.82	0.212	59.6	8.41e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0109	0.0127	0.15	0.173	0.181	0.196	9.52e-03	0.429	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.376	0.201	5.52	1.2	0.0595	5.41	0.167	0.159	0.353	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 57 | NRMSE (util.py)

T0 = 1412.0 K P0 = 4.083 atm phi0 = 0.677 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0428	0.666	8.62	0.853	0.176	0.956	0.0319	1.51	4.49	0	Complete
2 P-only DeepONet10	**10k**	0.0101	0.104	0.933	0.986	0.0348	0.33	0.0267	0.132	1.47	0	Complete
2 P-only DeepONet10	**long**	2.84e-03	0.323	0.857	0.952	5.85e-03	0.233	3.52e-03	0.0448	0.0253	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	5.19e-03	0.0455	0.906	1.27	0.0366	0.352	0.0232	0.0753	1.05	0	Complete
3 ANN10 + P-only DeepONet10	**long**	3.19e-03	0.298	0.87	1.07	0.0111	0.253	5.68e-03	0.0357	0.0136	0	Complete
4 T/P DeepONet10	**10k**	3.80e-03	0.0682	0.0852	0.0645	2.64e-03	0.0271	4.83e-03	0.0327	0.023	0	Complete
4 T/P DeepONet10	**long**	0.0326	0.181	0.466	0.189	0.0562	0.681	0.0453	0.47	0.442	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	4.76e-03	0.0786	0.0856	0.0623	1.90e-03	0.045	6.40e-03	0.0294	0.0218	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0188	0.081	0.233	0.0214	0.0372	0.455	0.0278	0.281	0.3	0	Complete
6 T/P/ϕ DeepONet10	**10k**	8.28e-03	0.0376	0.0669	0.036	0.03	0.288	0.0133	0.265	0.371	0	Complete
6 T/P/ϕ DeepONet10	**long**	5.07e-03	0.0825	0.107	0.0449	0.0148	0.143	8.31e-03	0.15	0.115	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	3.84e-03	0.0728	0.087	0.0721	0.0256	0.213	7.18e-03	0.213	0.325	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	2.88e-03	0.0541	0.0529	0.0535	0.0142	0.128	5.34e-03	0.115	0.134	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10
10k updates

2 P-only
DeepONet10

3 ANN10 +
P-only DeepONet10

4 T/P
DeepONet10

5 ANN10 +
T/P DeepONet10

6 T/P/ϕ
DeepONet10

7 ANN10 +
T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 64 | NRMSE (util.py)

T0 = 1045.3 K P0 = 4.301 atm phi0 = 0.595 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.284	0.791	1.02	1.04	0.509	11.4	0.523	1.02	1.01	0	Complete
2 P-only DeepONet10	**10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	**long**	0.336	0.848	1.02	1.81	0.455	0.553	0.622	1.05	1.43	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.294	0.742	0.98	1.61	0.399	0.504	0.545	1.18	1.79	0	Complete
4 T/P DeepONet10	**10k**	0.346	0.949	6.56	1.71	0.539	3.79	0.654	1.21	0.992	0	Complete
4 T/P DeepONet10	**long**	0.557	0.963	54.9	0.698	0.903	39.2	0.734	1.01	1	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.331	0.919	4.92	1.11	0.528	4.59	0.625	1.09	0.99	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.406	0.874	27.7	0.985	0.74	28.8	0.611	0.916	0.996	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.312	0.881	59.7	0.777	0.532	11.6	0.548	2.86	0.99	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 68 | NRMSE (util.py)

T0 = 1086.3 K P0 = 3.702 atm phi0 = 1.168 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.092	0.499	1.5	1.33	0.738	0.979	0.157	1.47	1.33	0	Complete
2 P-only DeepONet10	**10k**	0.615	3.15	1	1	4.94	1	1	0.986	0.989	0	Complete
2 P-only DeepONet10	**long**	0.248	0.947	0.947	9.35	0.916	2.28	0.352	1.23	1.07	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.0874	0.309	0.756	0.913	0.368	1.32	0.126	1.17	1.26	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0768	0.403	0.258	0.249	0.623	0.15	0.127	1.26	1.15	0	Complete
4 T/P DeepONet10	**10k**	0.124	0.51	1.4	0.786	0.778	1.36	0.174	2.17	1.35	0	Complete
4 T/P DeepONet10	**long**	0.48	0.362	11.2	0.253	0.283	5.35	0.284	0.924	1.07	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0776	0.325	1.01	0.993	0.528	1.02	0.102	1.28	1.09	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.3	0.302	3.93	0.68	0.431	5.48	0.291	2.17	1.34	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.132	0.389	0.981	1.04	0.557	1.8	0.167	1.73	1.92	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.296	0.376	4.5	0.345	0.578	4.66	0.273	1.91	1.16	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.219	0.369	2.06	0.265	0.472	4.04	0.245	1.75	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.236	0.366	3.17	0.598	0.563	4.14	0.237	1.49	1.18	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 69 | NRMSE (util.py)

T0 = 935.0 K P0 = 2.045 atm phi0 = 0.523 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	9.88e-07	1.28e-06	1	185	1.53e-06	1	1	0.967	0.997	0	Complete
2 P-only DeepONet10	**10k**	2.83e-03	8.45e-03	9.87e+06	5.45e+07	6.11e-03	7.94e+07	1.15e+04	139	52.9	0	Complete
2 P-only DeepONet10	**long**	9.69e-07	1.29e-06	0.999	0.872	1.56e-06	0.979	0.997	0.994	1	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	9.70e-07	1.29e-06	1	0.492	1.55e-06	0.998	0.999	0.988	0.999	0	Complete
3 ANN10 + P-only DeepONet10	**long**	9.69e-07	1.29e-06	1	0.796	1.56e-06	0.995	0.997	0.992	1	0	Complete
4 T/P DeepONet10	**10k**	9.45e-07	1.27e-06	1	2.28	1.55e-06	0.952	0.963	0.997	1	0	Complete
4 T/P DeepONet10	**long**	3.65e-06	5.13e-06	0.853	1.79	6.38e-06	1.04	3.39	3.99	4.75	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	3.86e-04	5.04e-04	1	13	5.70e-04	0.999	8.88	94.2	1.33e+03	0	Complete
5 ANN10 + T/P DeepONet10	**long**	3.81e-05	4.88e-05	0.575	20.2	5.71e-05	7.21	31.3	29.7	57.5	0	Complete
6 T/P/ϕ DeepONet10	**10k**	2.78e-05	2.32e-05	87.3	109	1.84e-05	399	35.7	7.04	8.79	0	Complete
6 T/P/ϕ DeepONet10	**long**	9.71e-07	1.29e-06	0.505	0.95	1.56e-06	0.868	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	9.40e-06	8.94e-06	18.9	16.9	8.37e-06	52.1	11.3	4.03	5.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	1.03e-06	1.25e-06	156	0.854	1.56e-06	3.06	0.983	0.994	1	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 73 | NRMSE (util.py)

T0 = 1106.0 K P0 = 1.911 atm phi0 = 1.915 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0381	0.0487	1.14	0.997	0.157	0.82	0.0187	0.373	6.14	0	Complete
2 P-only DeepONet10	**10k**	0.0376	0.025	0.757	0.734	0.154	0.77	0.019	0.37	0.648	0	Complete
2 P-only DeepONet10	**long**	3.35e-03	0.0147	0.0996	0.287	0.133	0.0551	0.0102	0.313	0.367	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.03	0.0143	0.592	0.58	7.95e-03	0.633	0.0124	0.0794	0.188	0	Complete
3 ANN10 + P-only DeepONet10	**long**	4.06e-03	0.0152	0.103	0.297	0.14	0.0617	0.0108	0.321	0.367	0	Complete
4 T/P DeepONet10	**10k**	0.0466	0.0258	0.956	0.772	0.0161	0.986	0.0171	0.459	6.48	0	Complete
4 T/P DeepONet10	**long**	0.115	0.0565	2.29	2.96	0.0407	2.17	0.0443	0.14	0.326	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0444	0.0276	0.93	0.673	0.0709	0.961	0.0159	0.422	2.4	0	Complete
5 ANN10 + T/P DeepONet10	**long**	4.01e-03	6.44e-03	0.0809	0.135	0.0572	0.0809	4.74e-03	0.148	0.127	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0497	0.0333	0.96	1.13	0.242	1.05	0.0279	0.634	0.879	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.247	0.141	5.03	4.37	0.116	4.45	0.0868	8.52	1.37e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0248	0.0245	0.472	0.561	0.21	0.58	0.0195	0.528	0.672	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.0744	0.0454	1.45	1.63	0.1	2.52	0.0462	0.284	28.5	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 74 | NRMSE (util.py)

T0 = 1222.9 K P0 = 4.971 atm phi0 = 1.766 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0331	0.0696	1.38	0.728	0.0408	0.804	0.0201	0.211	1.99	0	Complete
2 P-only DeepONet10	**10k**	0.0501	0.0284	1.01	1.17	0.177	1.1	0.0287	0.692	1.41	0	Complete
2 P-only DeepONet10	**long**	0.233	0.365	0.968	34.6	2.2	10.9	0.375	0.164	0.474	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.045	0.0562	1.06	2.06	0.212	1.67	0.0447	0.662	3.9	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.136	0.238	0.942	17.3	1.21	7.6	0.241	0.18	0.743	0	Complete
4 T/P DeepONet10	**10k**	0.04	0.0532	0.97	1.09	0.72	1	0.0224	1.15	9.51	0	Complete
4 T/P DeepONet10	**long**	0.232	0.221	6.11	0.839	0.37	2.14	0.0482	2.15	2.41	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	8.40e-03	0.0292	0.178	0.524	0.521	0.199	0.0177	1.07	0.981	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.175	0.176	4.73	0.924	0.338	1.41	0.032	1.24	1.45	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.155	0.0323	2.06	2	0.572	5.48	0.131	1.5	5.07	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.272	0.0958	4.94	0.121	0.0258	7.13	0.162	1.43	1.85	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.152	0.0322	1.86	0.606	0.517	5.99	0.138	1.4	6.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.211	0.058	3.61	0.397	0.0235	6.14	0.138	0.499	0.509	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

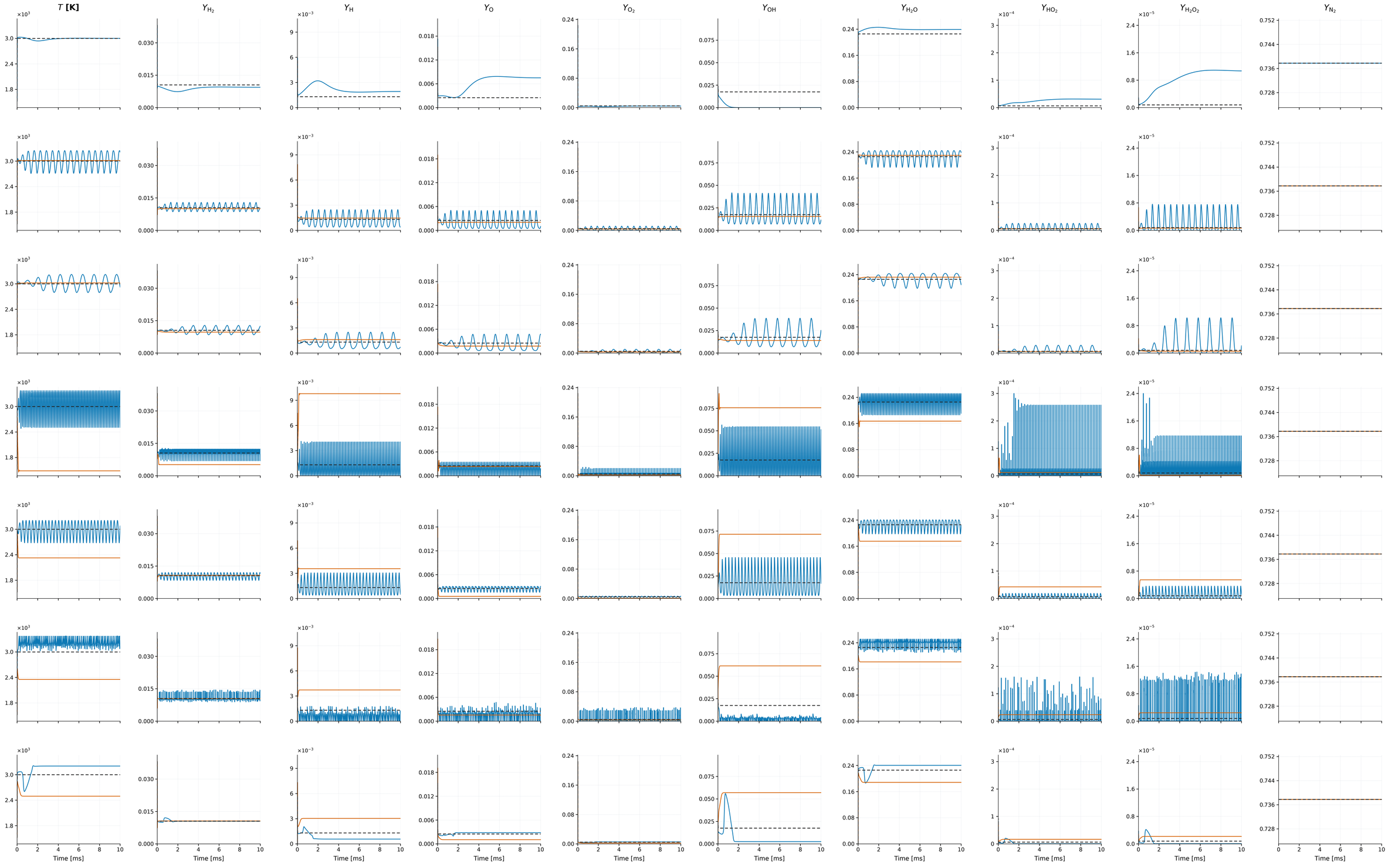
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 76 | NRMSE (util.py)

T0 = 1523.6 K P0 = 3.626 atm phi0 = 1.349 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	7.28e-03	0.157	0.733	1.66	0.164	0.967	0.0661	3.08	10.3	0	Complete
2 P-only DeepONet10	**10k**	0.0606	0.124	0.537	0.631	0.585	0.637	0.075	1.43	3.56	0	Complete
2 P-only DeepONet10	**long**	5.46e-03	0.0458	0.124	0.194	0.173	0.117	0.0176	0.174	0.215	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.0447	0.119	0.469	0.535	0.487	0.537	0.062	1.43	4.32	0	Complete
3 ANN10 + P-only DeepONet10	**long**	7.37e-03	0.0772	0.224	0.293	0.27	0.199	0.0288	0.178	0.333	0	Complete
4 T/P DeepONet10	**10k**	0.101	0.138	1.02	0.564	0.519	1.13	0.102	4.47	2.83	0	Complete
4 T/P DeepONet10	**long**	0.502	0.507	6.45	0.0877	0.396	3.32	0.259	1.13	0.876	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0593	0.113	0.713	0.206	0.165	0.838	0.067	0.92	1.56	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.224	0.0196	1.71	0.778	0.613	3.06	0.222	5.46	5.78	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.0794	0.0678	0.632	0.485	0.728	0.848	0.0779	2.08	2.75	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.215	0.0118	1.85	0.378	0.196	2.51	0.195	2.63	2.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0672	0.0325	0.52	0.132	0.106	0.894	0.0693	0.916	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.167	0.0121	1.31	0.565	0.383	2.23	0.164	1.54	1.7	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10
10k updates

2 P-only
DeepONet10

3 ANN10 +
P-only DeepONet10

4 T/P
DeepONet10

10k stopped: 1.084 ms

5 ANN10 +
T/P DeepONet10

6 T/P/ϕ
DeepONet10

7 ANN10 +
T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 78 | NRMSE (util.py)

T0 = 962.3 K P0 = 2.443 atm phi0 = 1.140 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	4.72e-05	2.63e-05	3.24	4.29e+03	6.52e-05	1	1	1.07	2.06	0	Complete
2 P-only DeepONet10	**10k**	3.54e-05	1.95e-05	1	0.876	4.00e-05	0.999	0.869	0.848	0.718	0	Complete
2 P-only DeepONet10	**long**	0.91	0.519	7.83e+04	5.98e+07	0.743	2.52e+08	3.49e+04	5.36	48.8	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.16	0.0496	8.47e+05	6.63e+07	0.166	1.05e+08	1.77e+03	16.5	4.06	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.516	0.234	4.10e+04	7.65e+06	0.288	3.86e+07	1.73e+04	11.6	103	0	Complete
4 T/P DeepONet10	**10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.084 ms**
4 T/P DeepONet10	**long**	4.54e-05	2.58e-05	0.982	0.991	5.41e-05	0.992	0.998	0.992	0.999	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	1.86	0.777	1.64e+06	1.13e+06	0.886	3.31e+07	5.98e+04	8.67	37.4	0	Complete
5 ANN10 + T/P DeepONet10	**long**	6.30e-04	2.92e-04	22.3	11	4.71e-04	17.9	15.1	2.83	10.3	0	Complete
6 T/P/ϕ DeepONet10	**10k**	4.54e-05	2.59e-05	1	0.884	5.42e-05	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	5.35e-05	2.92e-05	2.59	1.21	5.90e-05	1.9	1.23	1.02	1.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	4.54e-05	2.58e-05	1	0.819	5.42e-05	0.995	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.473	0.575	1.67e+07	2.70e+06	0.642	3.16e+08	3.03e+04	8.6	36.9	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.

Cantera vs neural networks | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference NN: 10,000 updates NN: long training (time-limited)

1 ANN10 10k updates

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scale within each variable column. All curves use saved 2 us samples. Failed predictions stop at their last accepted state (x).
ANN always uses its 10k checkpoint; mixed teams use ANN10 + DeepONet10 at the same current state. Long endpoints reached their wall-time caps, without verified loss plateaus.

Cantera vs NN | Test IC 88 | NRMSE (util.py)

T0 = 1502.1 K P0 = 4.980 atm phi0 = 0.859 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0536	0.764	5.02	0.609	0.392	0.914	0.0409	1.27	4.48	0	Complete
2 P-only DeepONet10	**10k**	0.101	0.92	0.803	0.75	0.101	1.19	0.157	5.76	17.3	0	Complete
2 P-only DeepONet10	**long**	0.034	0.773	0.87	1.37	0.128	0.0762	0.058	0.0685	0.482	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.234	0.143	2.9	0.928	0.832	4.28	0.3	1.4	7.8	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0197	0.591	0.854	1.14	0.0875	0.122	0.0369	0.0377	0.426	0	Complete
4 T/P DeepONet10	**10k**	0.0871	0.306	1.15	0.406	0.106	0.998	0.107	2.11	2.76	0	Complete
4 T/P DeepONet10	**long**	0.0318	0.0936	0.327	0.0664	0.0608	0.439	0.0414	0.372	0.308	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0113	0.0567	0.186	0.0697	0.0111	0.105	0.0136	0.111	0.106	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0209	0.0376	0.204	0.0232	0.059	0.35	0.0283	0.233	0.213	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.024	0.107	0.479	0.124	0.0122	0.217	0.0275	0.295	0.232	0	Complete
6 T/P/ϕ DeepONet10	**long**	8.43e-03	0.0801	0.106	0.0407	0.0381	0.112	0.0126	0.0744	0.0497	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0144	0.076	0.19	0.0787	0.012	0.145	0.0184	0.122	0.129	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	6.41e-03	0.0409	0.0796	0.0151	0.0166	0.0767	8.86e-03	0.033	0.0258	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.